CHAPTER 03

Analysis of Linear Programming Models - Part 3

Equality constraints and special cases: infeasible, unbounded, and multiple optima

TEXTBOOK	CHAPTER	WEEK
Modern Management Science, 9th ed.	Ch. 3	Week 4

Management Science · Fall 2026 · Kitae Kim, School of Business, Korea Aerospace University

Content prepared by the instructor; slide layout and formatting assisted by Claude (Anthropic).

Analysis with software

- Google Colab: a free, cloud-based Python environment
- CVXPY: write the model the way the math reads, then hand it to a solver
- Many solvers, one interface: choose a solver that suits the problem class and size
- LP problems solved in six steps:
    1. variables,
    2. objective,
    3. constraints,
    4. problem,
    5. solve,
    6. results

PART 3

Equality Constraints and Special Cases

What the solver tells you when a model has no ordinary optimum

- Equality constraints (Example 3-3)
- Infeasible models (Example 3-4)
- Unbounded models (Example 3-5)
- Multiple optimal solutions (Example 3-1)

Example 3-3: an equality constraint

PROBLEM

K Electronics plans to hire skilled workers and apprentices as it expands into a new line of business. A skilled worker costs ₩10,000 per hour and an apprentice ₩5,000 per hour. The company has ₩150 million available for the new hiring plan, and under its agreement with the labor union this entire amount must be spent on hiring. Given its current situation, K Electronics plans to hire skilled workers for no more than 10,000 hours, while the union demands that apprentices be hired for at least 12,000 hours. Current cost analysis shows that a skilled worker earns the company ₩3,000 per hour and an apprentice ₩1,250 per hour. Management wants a hiring plan that maximizes total profit.

FORMULATION

- **Decision variables**
- **Objective function**
- **Constraints**

Example 3-3: from model to graph

FORMULATION

- **Decision variables**
  - S = hours of skilled workers hired
  - P = hours of apprentices hired
- **Objective function**
  - **Maximize** Z = 0.3S + 0.125P (profit, unit: ₩10,000)
- **Constraints**
  - S + 0.5P = 15,000 (budget, unit: ₩10,000)
  - S ≤ 10,000 (skilled hours, unit: hours)
  - P ≥ 12,000 (apprentice hours, unit: hours)
  - S, P ≥ 0

GRAPH

Example 3-3: input and output

INPUT

```
import cvxpy as cp

# 1. Decision variables
S = cp.Variable()
P = cp.Variable()

# 2. Objective: maximize total profit
objective = cp.Maximize(0.3 * S + 0.125 * P)

# 3. Constraints (note the == for equality)
constraints = [
    S + 0.5 * P == 15000,  # budget: spend all of it
    S <= 10000,            # skilled-worker hours
    P >= 12000,            # apprentice hours
    S >= 0,                # non-negativity
    P >= 0
]

# 4. Formulate and solve the problem
problem = cp.Problem(objective, constraints)
problem.solve()

# 5. Print the results
print("Status:", problem.status)
print("Maximum profit (Z):", problem.value)
print("Optimal S:", S.value)
print("Optimal P:", P.value)
```

OUTPUT

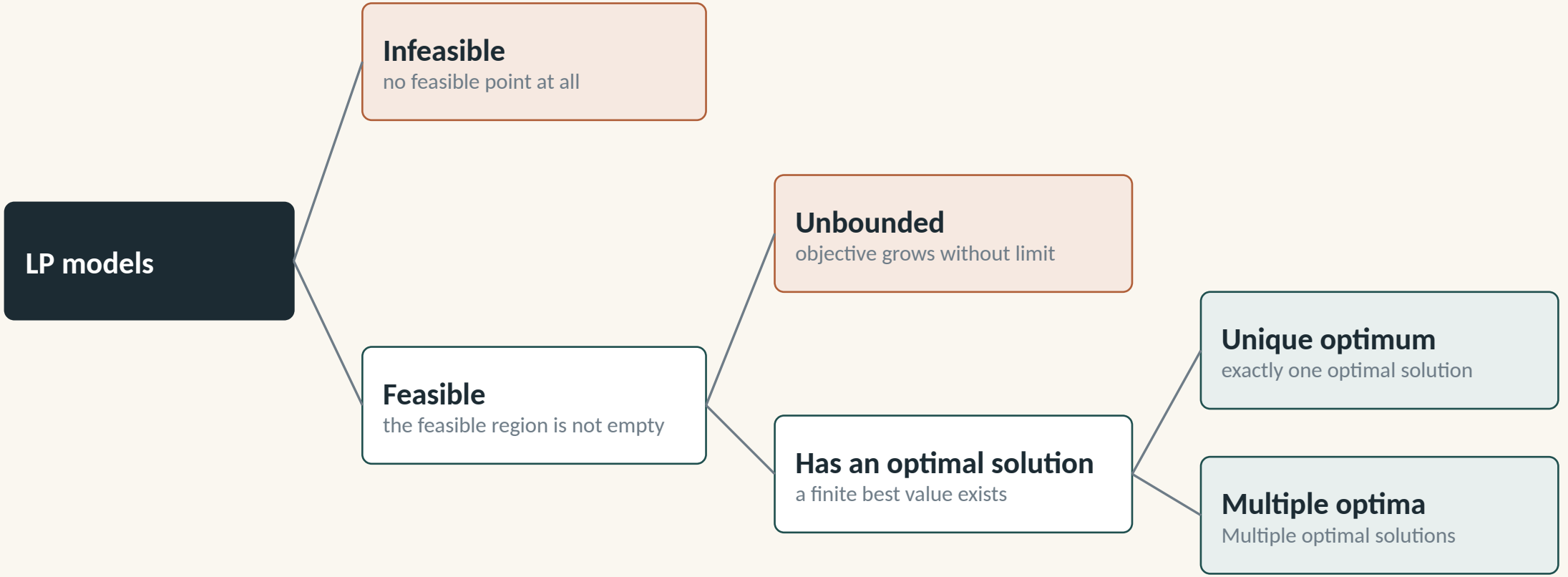
```
Status: optimal
Maximum profit (Z): 4199.999942204862
Optimal S: 8999.99884409724
Optimal P: 12000.002311805523
```

- Status is optimal: the equality leaves only a line segment, but a best point still exists
- Hire skilled workers for 9,000 hours and apprentices for 12,000 hours
- The whole budget is spent: 9,000 + 0.5 × 12,000 = 15,000 (₩10,000s)

Maximum profit: ₩42 million (Z = 4,200 in ₩10,000s)

SPECIAL CASES

Not every LP model has an optimal solution



EXAMPLE 3-4

Example 3-4: an infeasible model

PROBLEM

Architect Park has been commissioned by the government to design an affordable solar-energy house. The house consists of a residential area and an energy-control area: the residential area must be at least 105 m^2, and the energy-control area at least 30 m^2. Construction costs ₩2 million per m^2 for the residential area and ₩4 million per m^2 for the energy-control area. The government requires a design that maximizes total floor area within a total construction budget of ₩240 million.

FORMULATION

- **Decision variables**
- **Objective function**
- **Constraints**

Example 3-4: from model to graph

FORMULATION

- **Decision variables**
  - R = residential area (m^2)
  - E = energy-control area (m^2)
- **Objective function**
  - **Maximize** $Z = R + E$ (total area, m^2)
- **Constraints**
  - $200R + 400E \leq 24{,}000$ (cost, ₩10,000s)
  - $R \geq 105$ (residential area, m^2)
  - $E \geq 30$ (energy-control area, m^2)
  - $R, E \geq 0$

GRAPH

Example 3-4: output

INPUT

```
import cvxpy as cp

# 1. Decision variables
R = cp.Variable()
E = cp.Variable()

# 2. Objective: maximize total floor area
objective = cp.Maximize(R + E)

# 3. Constraints
constraints = [
    200 * R + 400 * E <= 24000,  # construction cost
    R >= 105,                    # residential area
    E >= 30,                     # energy-control area
    R >= 0,                      # non-negativity
    E >= 0
]

# 4. Formulate and solve the problem
problem = cp.Problem(objective, constraints)
problem.solve()

# 5. Print the results
print("Status:", problem.status)
print("Maximum total area:", problem.value)
print("Optimal R:", R.value)
print("Optimal E:", E.value)
```

OUTPUT

```
Status: infeasible
Maximum total area: -inf
Optimal R: None
Optimal E: None
```

- Status is infeasible: no point satisfies all the constraints at once
- The minimum areas alone cost 200 × 105 + 400 × 30 = 33,000, above the 24,000 budget
- There is no solution to report: the variables are None and the objective is −inf

Infeasible: relax a requirement or raise the budget

Example 3-5: an unbounded model

PROBLEM

S Industries produces two special-purpose motors, type A and type B. To meet next month's supply contracts, at least 1,000 units of each type must be produced. As a matter of company policy, the production volume of Type B motors should not exceed that of Type A motors by more than 500 units. The profit per unit is ₩200,000 for type A and ₩100,000 for type B. S Industries wants a production plan that maximizes total profit.

FORMULATION

- **Decision variables**
- **Objective function**
- **Constraints**

Example 3-5: from model to graph

FORMULATION

- **Decision variables**
  - X_1 = units of type A motors produced
  - X_2 = units of type B motors produced
- **Objective function**
  - **Maximize** $Z = 20X_1 + 10X_2$ (profit, ₩10,000s)
- **Constraints**
  - $X_1 \geq 1{,}000$ (type A contract)
  - $X_2 \geq 1{,}000$ (type B contract)
  - $X_2 - X_1 \leq 500$ (company policy)
  - $X_1, X_2 \geq 0$

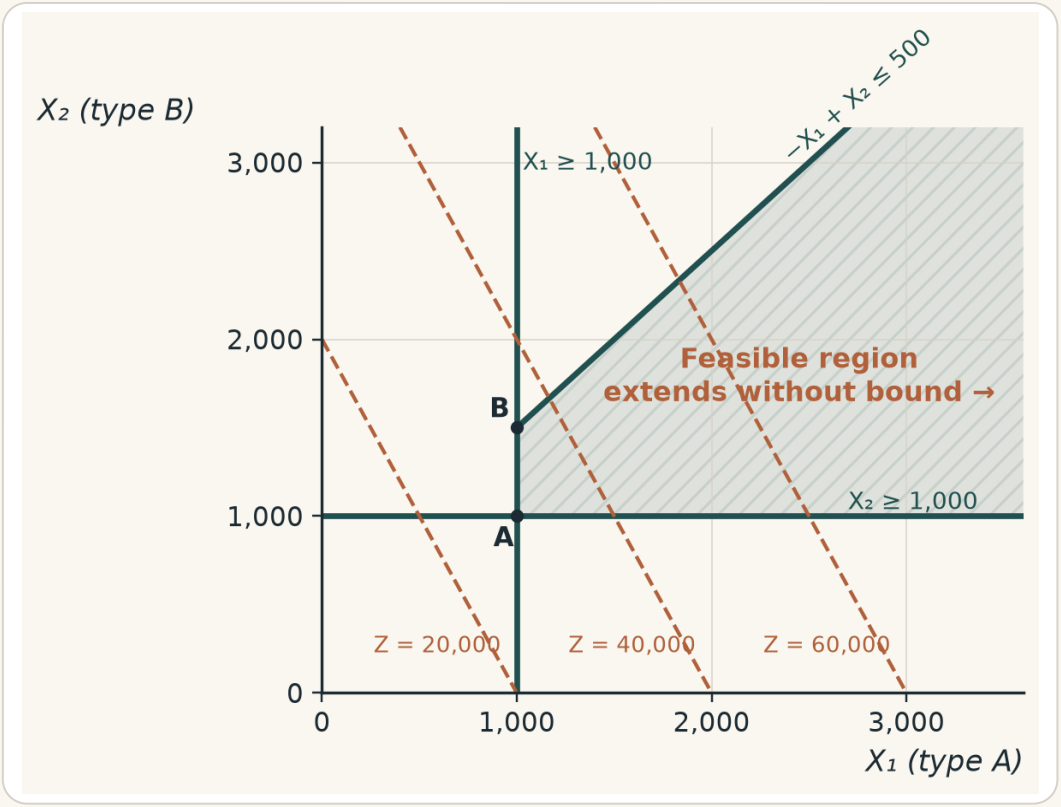
GRAPH

Example 3-5: from model to graph

FORMULATION

- **Decision variables**
  - X_1 = units of type A motors produced
  - X_2 = units of type B motors produced
- **Objective function**
  - **Maximize** $Z = 20X_1 + 10X_2$ (profit, ₩10,000s)
- **Constraints**
  - $X_1 \geq 1{,}000$ (type A contract)
  - $X_2 \geq 1{,}000$ (type B contract)
  - $X_2 - X_1 \leq 500$ (company policy)
  - $X_1, X_2 \geq 0$

GRAPH



Example 3-5: input to output

INPUT

```
import cvxpy as cp

# 1. Decision variables
X1 = cp.Variable()
X2 = cp.Variable()

# 2. Objective: maximize total profit
objective = cp.Maximize(20 * X1 + 10 * X2)

# 3. Constraints
constraints = [
    X1 >= 1000,          # type A supply contract
    X2 >= 1000,          # type B supply contract
    X2 - X1 <= 500,      # company policy
    X1 >= 0,             # non-negativity
    X2 >= 0
]

# 4. Formulate and solve the problem
problem = cp.Problem(objective, constraints)
problem.solve()

# 5. Print the results
print("Status:", problem.status)
print("Maximum total profit:", problem.value)
print("Optimal X1:", X1.value)
print("Optimal X2:", X2.value)
```

OUTPUT

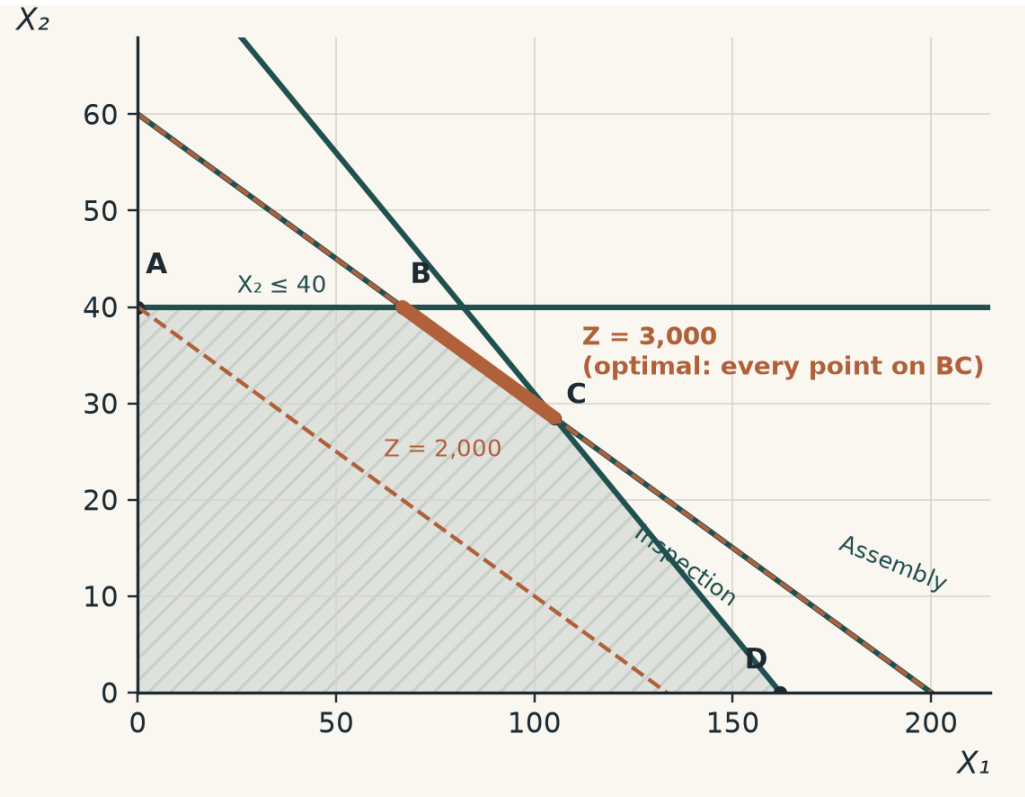
```
Status: unbounded
Maximum total profit: inf
Optimal X1: None
Optimal X2: None
```

- Status is unbounded: the feasible region has no upper limit, so profit grows without bound
- No finite optimum exists, so the variables have no value (None)
- In practice this signals a missing constraint, such as a capacity or demand limit

Unbounded: a constraint is missing from the model

Multiple optimal solutions

- Some LP models have more than one optimal solution
- Example 3-1 with a modified objective function
  - **Maximize** $Z = 15X_1 + 50X_2$ (was $20X_1 + 50X_2$)
  - $1.2X_1 + 4X_2 \leq 240$ (assembly, hours)
  - $0.5X_1 + X_2 \leq 81$ (inspection, hours)
  - $X_2 \leq 40$ (chip supply, units)
  - $X_1, X_2 \geq 0$
- The objective line is now parallel to the assembly constraint, so every point on segment BC is optimal (Z = 3,000)



Try it yourself: [interactive LP simulator (click to open)]

Summary

- What the solver's status tells you
  - optimal: a finite best solution exists (Ex. 3-3)
  - infeasible: the constraints contradict one another (Ex. 3-4)
  - unbounded: the objective grows without limit (Ex. 3-5)
- Equality constraints are written with == in CVXPY
- Multiple optima arise when the objective is parallel to a binding constraint

STATUS CHEAT SHEET

optimal
`value is finite; X.value is set`

infeasible
`value = −inf (max) or inf (min); X.value = None`

unbounded
`value = inf (max) or −inf (min); X.value = None`

equality constraint
```
S + 0.5 * P == 15000
```

next class
`post-optimality and sensitivity analysis`